

Dmytro Palahin

Quantitative Engineer / Quantitative Developer

📍 Paris, France

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Final-year Computer Science engineering student with **3 years of experience in Data Engineering, Machine Learning and MLOps** at **Société Générale Assurances**. Strong background in **applied mathematics, probability, statistics, optimization and algorithms**, complemented by hands-on experience with **Python, C/C++, scientific computing and large-scale data processing**.

🏢 PROFESSIONAL EXPERIENCE

Data Engineer / MLOps (Work-study) — Société Générale Assurances

09/2023 – Present

La Défense, France

- Analyzed **1 year of operational data** from the **Zaion Callbot** using Python to identify the main sources of errors, assess system performance, and formulate **3 improvement recommendations**
– Python · Pandas · NumPy · Jupyter Notebook · Matplotlib · Plotly
- Designed an **automated SAS-to-Python migration system**, integrating data processing with **Polars and DuckDB**, prompt engineering, and **automated validation through a 2-level test suite**
– Python · Polars · DuckDB · GitHub Copilot · AWS S3
- Designed and automated a **data processing and migration pipeline** enabling the transfer of **500+ GB of SAS datasets to AWS S3** and making them available to Data teams
– Python · AWS S3 · fsspec · smbfs · Linux · Bash

🔬 TECHNICAL & SCIENTIFIC PROJECTS

- Combinatorial Optimization of Logistics Centers** | Python, Julia, CPLEX, MILP **2025**
Modeling and solving a facility location problem using **mixed-integer linear programming**, CPLEX and heuristic methods
- Scientific Publication — Mathematical Algorithms** **2021**
Development of a mathematical method for impulse noise filtering in video images, published in *Informatics and Mathematical Methods in Simulation*, Vol. 11, No. 4
- Anomaly Detection** | Python, Machine Learning **2021**
Study of anomaly detection methods and partial data labeling
- Traveling Salesman Problem** | C, Combinatorial Optimization, Metaheuristics **2023**
Implementation in C of an optimization algorithm based on **ant colonies**
- Machine Learning Phishing Detection** | Python, Machine Learning, NLP **2026**
Classification of **76 677 emails** from 5 real-world datasets using TF-IDF, LightGBM, Naive Bayes et LSTM

🎓 EDUCATION

Sup Galilée Engineering School — Engineering Degree in Computer Science (*in progress*)

2023 – 2026

Courses: Probability and Statistics, Linear and Combinatorial Optimization, Numerical Methods, Data Mining, Programming for AI, Algorithms and Complexity, Graph Algorithms, C/C++, Parallel and Distributed Algorithms, Advanced Databases, Cloud Computing

Sorbonne Paris Nord University — Preparatory Class for Sup Galilée

2021 – 2023

Courses: Linear Algebra (vector spaces, matrices, eigenvalues, diagonalization), Multivariable Analysis (partial derivatives, Jacobian, Hessian, Taylor series), Probability (random variables, conditional probability, common distributions), differential equations and numerical computing

📜 CERTIFICATIONS & SKILLS

- Programming & Scientific Computing:** C/C++, Python, SQL, Java · NumPy, Pandas, Polars, DuckDB · Git, Linux, Jupyter
- Algorithms & Computer Science:** Data structures, algorithmic complexity, graph algorithms, parallel and distributed programming
- Applied Mathematics:** Linear algebra, multivariable analysis, probability and statistics, numerical methods, linear and combinatorial optimization, differential equations
- Machine Learning:** Scikit-Learn, LightGBM, XGBoost, TensorFlow/Keras · Data Mining, classification, anomaly detection
- Financial Markets & Market Data:** Forex, equities, futures · Market microstructure, order book (L2/L3), market data, liquidity, order types, execution, VWAP · COT reports (Commitments of Traders), risk management
- Udemy:** *Python for Data Science and Machine Learning Bootcamp* — Jose Portilla / Pierian Training

🏆 AWARDS & PARTICIPATIONS

- Georges Besse Foundation Laureate**
- Eco – Techno Ukraine ISEF** — 1st place in Computer Science & **Collective Intelligence Internet Service** — 1st place in Ukraine

🗨️ LANGUAGES & INTERESTS

French: Fluent **English:** Fluent **Ukrainian:** Native **Russian:** Bilingual

📊 Financial Markets / Trading ♟️ Chess ⛷️ Ski 🎾 Tennis 🏊 Swimming 🏎️ Karting | Formula 1